Chapter 2. Distribution-Free Inference for One Population

Sign test, Wilcoxon signed-rank test, Hodges–Lehmann estimator, efficiency comparisons

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Setting

$X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} F$ with median θ. We test $H_0 : \theta = \theta_0$ (against $\theta > \theta_0$, $\theta < \theta_0$ or $\theta \neq \theta_0$), estimate θ, and give confidence intervals, under two assumptions of increasing strength: F continuous (sign test), and F continuous and symmetric about θ (signed-rank test).

2.1 The sign test

Motivation. If we know nothing about the distribution except that it is continuous, the number of observations above the hypothesized median is still exactly binomial. The sign test uses only this, so it is valid for any distribution.

Definition (Sign statistic)

$S = S(\theta_0) = \sum_{i=1}^{n} I(X_i > \theta_0)$, the number of observations above θ_0.

Proposition 2.1

If $P(X_i = \theta_0) = 0$ and θ_0 is the median of F, then $S \sim B(n, \frac{1}{2})$ for every such F.

Proof. The indicators $I(X_i > \theta_0)$ are i.i.d. Bernoulli with success probability $P(X_1 > \theta_0) = 1 - F(\theta_0) = \frac{1}{2}$. ■

Thus the sign test is distribution-free. Under $\theta > \theta_0$ each sign is "+" with probability $p_1 = P(X > \theta_0) > \frac{1}{2}$, so we reject for large S. The upper-tailed p-value is $P(S \geq s_{\text{obs}})$; the two-sided one is $2\min\{P(S \geq s_{\text{obs}}), P(S \leq s_{\text{obs}})\}$, using the symmetry of $B(n, \frac{1}{2})$.

Proposition 2.2 (Rejection region)

The smallest c with $P(S \geq c) \leq \alpha$ under H_0 is $c = Q_B(1-\alpha) + 1$, where Q_B is the quantile function of $B(n, \frac{1}{2})$.

Proof. $P(S \geq c) \leq \alpha \iff F_B(c-1) \geq 1 - \alpha$, and the smallest integer $c - 1$ with this property is $Q_B(1-\alpha)$ (Proposition 1.1). ■

Because S is discrete, the attained size is usually below α. For large n, $(S - n/2)/\sqrt{n/4} \dot{\sim} N(0,1)$.

Ties. Observations equal to θ_0 are discarded and n is reduced accordingly; conditionally on the number discarded, the remaining signs are again i.i.d. fair coins whenever $P(X < \theta_0) = P(X > \theta_0)$.

Example 2.1

Twelve paired differences (after − before) have 9 positive signs. For $H_1 : \theta > 0$, $p = P(S \geq 9) = (220 + 66 + 12 + 1)/4096 = 0.073$.

Power. If the alternative population is known, $S \sim B(n, p_1)$ with $p_1 = P(X > \theta_0)$, so the power is $1 - F_B(c - 1; n, p_1)$. It depends on the shape of F, not only on θ.

Example 2.2 (Sign test vs. z-test)

Let $n = 20$, $\theta_0 = 0$, $\theta = 0.5$, standard deviation 1. The size-≤ 0.05 sign test rejects when $S \geq 15$ (actual size 0.021).

- Normal $N(0.5, 1)$: $p_1 = \Phi(0.5) = 0.69$, power 0.38; the z-test has power $\Phi(0.5\sqrt{20} - 1.645) = 0.72$.
- Laplace with the same mean and variance (scale $1/\sqrt{2}$): $p_1 = 1 - \frac{1}{2}e^{-0.5\sqrt{2}} = 0.75$, power 0.63. The z-test still has power about 0.72.

The sign test gains ground when the tails are heavy. The asymptotic comparison is made precise in Section 2.3.

2.1.1 Confidence interval for the median

Proposition 2.3

Let F be continuous, $X_{(1)} < \cdots < X_{(n)}$ the order statistics and $l = Q_B(\alpha/2)$, $u = n - l + 1$. Then

$$P(X_{(l)} \leq \theta < X_{(u)}) = \sum_{k=l}^{n-l} \binom{n}{k} 2^{-n} \geq 1 - \alpha.$$

Proof. $X_{(l)} \leq \theta$ iff at most $n-l$ observations exceed θ, and $\theta < X_{(u)}$ iff at least $n-u+1 = l$ do. So the event is $\{l \leq S(\theta) \leq n-l\}$ with $S(\theta) \sim B(n, \frac{1}{2})$. By symmetry $P(S \leq l-1) = P(S \geq n-l+1)$, and each is $\leq \alpha/2$ by the choice of l (if $P(S \leq Q_B(\alpha/2)) = \alpha/2$ exactly, take $l = Q_B(\alpha/2) + 1$). ■

For $n = 15$ and $\alpha = 0.05$: $l = 4$, and $(X_{(4)}, X_{(12)})$ has coverage 0.965. For large n, the normal approximation $S \dot{\sim} N(n/2, n/4)$ gives the orders

$$l \approx \frac{n}{2} - z_{\alpha/2}\frac{\sqrt{n}}{2}, \qquad u \approx \frac{n}{2} + z_{\alpha/2}\frac{\sqrt{n}}{2} + 1 \qquad \text{(rounded to integers)},$$

which still yields an asymptotically valid, distribution-free interval. As the level decreases the interval shrinks to the sample median, the estimator associated with the sign test.

The same argument handles any quantile $Q(p)$: $\#\{X_i \leq Q(p)\} \sim B(n, p)$, so $(X_{(l)}, X_{(u)})$ covers $Q(p)$ with probability $\sum_{k=l}^{u-1} \binom{n}{k} p^k (1-p)^{n-k}$.

2.2 The Wilcoxon signed-rank test

Motivation. The sign test throws away magnitudes. If the distribution is symmetric, the ranks of $|X_i|$ can be used too; the signed-rank statistic remains distribution-free and is almost as efficient as the t-test under normality.

Assume now that F is continuous and symmetric about θ. Let $D_i = X_i - \theta_0$, $V_i = I(D_i > 0)$, and let R_i be the rank of $|D_i|$ among $|D_1|, \ldots, |D_n|$.

Definition (Signed-rank statistic)

$SR_+ = \sum_{i=1}^{n} V_i R_i$, the sum of the ranks of the positive deviations.

Lemma 2.4

Under H_0 (symmetry about θ_0), $(V_1, \ldots, V_n)$ are i.i.d. Bernoulli($\frac{1}{2}$) and independent of $(|D_1|, \ldots, |D_n|)$. Consequently

$$SR_+ \overset{d}{=} \sum_{k=1}^{n} kB_k, \qquad B_k \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(\tfrac{1}{2}).$$

Proof. For a symmetric continuous D, $\text{sgn}(D)$ and $|D|$ are independent with $P(D > 0) = \frac{1}{2}$: $P(D > 0, |D| \le t) = P(0 < D \le t) = \frac{1}{2}P(|D| \le t)$. Independence across i gives the first claim. Given the ranks, (R_i) is a permutation π of $(1, \ldots, n)$ and $\sum_i V_i R_i = \sum_k kV_{\pi^{-1}(k)}$, with $(V_{\pi^{-1}(k)})_k$ again i.i.d. fair coins. ■

Proposition 2.5

Under H_0, the null distribution of SR_+ depends only on n, is symmetric about $n(n+1)/4$, and

$$E(SR_+) = \frac{n(n+1)}{4}, \qquad \text{Var}(SR_+) = \frac{n(n+1)(2n+1)}{24}, \qquad \frac{SR_+ - E(SR_+)}{\sqrt{\text{Var}(SR_+)}} \xrightarrow{d} N(0,1).$$

Proof [Supplement]. From Lemma 2.4, $E = \frac{1}{2}\sum k$ and $\text{Var} = \frac{1}{4}\sum k^2$. Replacing B_k by $1 - B_k$ (same law) maps SR_+ to $\frac{n(n+1)}{2} - SR_+$, giving symmetry. For normality use Lyapunov's CLT with $U_k = kB_k$: $\sum_k E|U_k - k/2|^3 = \sum k^3/8 = O(n^4)$, while $s_n^3 = \text{Var}(SR_+)^{3/2} \asymp n^{9/2}$, so the Lyapunov ratio is $O(n^{-1/2}) \to 0$. ■

The exact null distribution is obtained by enumerating the 2^n equally likely sign patterns of the ranks (the *sign-flip* or re-randomization principle). The p-values are $P(SR_+ \ge sr_{\text{obs}})$ against $\theta > \theta_0$ and $P(SR_+ \le sr_{\text{obs}})$ against $\theta < \theta_0$. For the two-sided alternative, the p-value is twice the smaller one, which is legitimate here because the null distribution is symmetric.

Example 2.3

Deviations -0.3, 0.8, 1.5, 2.2, 2.9 from θ_0. The absolute ranks are $1, \ldots, 5$ and only rank 1 is negative, so $SR_+ = 2 + 3 + 4 + 5 = 14$. Among the 32 sign patterns, $SR_+ \ge 14$ occurs for "all positive" (15) and "only rank 1 negative" (14), so the one-sided p-value is $2/32 = 0.0625$. The sign test gives $P(S \ge 4) = 6/32 = 0.19$: the ranks carry more information.

Ties. Zero deviations are dropped. Tied $|D_i|$ receive mid-ranks. The null distribution is then the sign-flip distribution of the actual (mid-)ranks, which no longer has the form in Lemma 2.4 but is still exact conditionally. For the normal approximation use $\text{Var}(SR_+) = \frac{1}{4}\sum_i R_i^2$.

2.2.1 Walsh averages, the Hodges–Lehmann estimator and intervals

The $N = n(n+1)/2$ **Walsh averages** are $W_{ij} = (X_i + X_j)/2$, $1 \le i \le j \le n$.

Proposition 2.6

If no $|X_k - \theta|$ are tied and no W_{ij} equals θ, then $SR_+(\theta) = \#\{(i \le j) : W_{ij} > \theta\}$.

Proof [Supplement]. Let $d_k = X_k - \theta$, so $W_{ij} > \theta \iff d_i + d_j > 0$. Assign each pair $\{i, j\}$ to the index $k \in \{i, j\}$ with larger $|d_k|$. The sign of $d_i + d_j$ is the sign of d_k. Index k receives exactly the pairs $\{k, m\}$ with $|d_m| \le |d_k|$, i.e. R_k pairs. Hence $\#\{W_{ij} > \theta\} = \sum_{k:d_k>0} R_k$. ■

So the signed-rank statistic is the sign statistic applied to the Walsh averages, and Proposition 2.3 transfers.

Corollary 2.7

Let $W_{(1)} < \cdots < W_{(N)}$ and $l = Q_{SR}(\alpha/2)$ (quantile of the null distribution of SR_+), $u = N - l + 1$. Then $P(W_{(l)} \le \theta < W_{(u)}) \ge 1 - \alpha$. The point estimator obtained as the level shrinks is the **Hodges–Lehmann estimator** $\hat{\theta}_{HL} = \text{median}_{i \le j} W_{ij}$.

Proof. $\theta < W_{(a)} \iff SR_+(\theta) > N - a$, and $\theta \ge W_{(a)} \iff SR_+(\theta) \le N - a$. At the true θ, $SR_+(\theta)$ has the null distribution, which is symmetric about $N/2$. Repeat the proof of Proposition 2.3. ■

Paired data are handled by applying these one-sample methods to the differences.

2.3 Comparing the t-, sign and signed-rank procedures

Validity. The sign test is exact for any continuous F, and the signed-rank test for any continuous symmetric F. The t-test is exact only under normality. It is approximately valid when the CLT applies, but can be badly conservative for very heavy tails (where $\sigma^2 = \infty$, e.g. Cauchy or slash) and anticonservative for skewed F in moderate samples. For asymmetric F, mean and median differ, so the t-test does not even test the median. Typical behavior of the nominal 5% one-sided t-test:

- light tails (uniform): size slightly above 5% for $n \approx 5$, essentially correct by $n \approx 10$;
- infinite variance (slash, Cauchy): size well below 5% (very conservative), and this does *not* improve as n grows, because the CLT fails;
- skewed (exponential, testing the mean): convergence is slow, and the size can still be noticeably above 5% at $n = 100$; a corrected critical value is needed to restore it;
- skewed, testing the median with the t-statistic: the statistic is centered at the wrong value and the test is meaningless.

A test whose size exceeds α cannot be compared on power. Valid tests (or tests with corrected critical values) are compared by power at plausible alternatives, which depends on F, n and α.

Efficiency. The relative efficiency of estimators is $RE(\hat{\theta}_1, \hat{\theta}_2) = \text{MSE}(\hat{\theta}_2)/\text{MSE}(\hat{\theta}_1)$, and the **asymptotic relative efficiency** (ARE) is its limit as $n \to \infty$. For tests, the (Pitman) ARE is the limiting ratio N_2/N_1 of the sample sizes needed for equal power at alternatives $\theta_n \to \theta_0$. For these three procedures it equals the ARE of the associated estimators $\bar{X}_n$, median X_i and $\hat{\theta}_{HL}$.

For confidence intervals with the same (exact or asymptotic) coverage, define the efficiency as $1/\text{length}^2$. If $\hat{\theta} \dot{\sim} N(\theta, v/n)$, the interval $\hat{\theta} \pm z_{\alpha/2}\sqrt{v/n}$ has length

$$2z_{\alpha/2}\sqrt{v/n} \approx 2z_{\alpha/2}\sqrt{\text{MSE}(\hat{\theta})}.$$

Hence the ARE of two such intervals is the limit of the ratio of the mean squared errors of the two estimators, i.e. the ARE of the estimators. So estimators, tests and intervals are ranked the same way. For a normal population, e.g., $ARE(\bar{X}_n, \text{median } X_i) = \pi/2 \approx 1.57$.

Proposition 2.8

Let F have density f symmetric about θ, with $f(\theta) > 0$ and variance σ^2 (possibly infinite). Then

$$\sqrt{n}(\bar{X}_n - \theta) \xrightarrow{d} N(0, \sigma^2), \qquad \sqrt{n}(\text{median } X_i - \theta) \xrightarrow{d} N\Big(0, \frac{1}{4f(\theta)^2}\Big),$$
$$\sqrt{n}(\hat{\theta}_{HL} - \theta) \xrightarrow{d} N\Big(0, \frac{1}{12(\int f^2)^2}\Big).$$

Hence

$$ARE(\text{sign}, t) = 4\sigma^2 f(\theta)^2, \qquad ARE(SR, t) = 12\sigma^2\Big(\int f^2\Big)^2, \qquad ARE(\text{sign}, SR) = \frac{f(\theta)^2}{3(\int f^2)^2}.$$

Proof for the median; sketch for HL. $P(\text{median } X_i \le \theta + t/\sqrt{n}) = P(\#\{X_i \le \theta + t/\sqrt{n}\} \ge \lceil n/2 \rceil)$. The count is $B(n, p_n)$ with $p_n = \frac{1}{2} + f(\theta)t/\sqrt{n} + o(n^{-1/2})$. The normal approximation of the binomial gives the limit $\Phi(2f(\theta)t)$, i.e. variance $1/(4f(\theta)^2)$. For $\hat{\theta}_{HL}$ argue the same way with Proposition 2.6: $\hat{\theta}_{HL} \le \theta + t/\sqrt{n}$ iff $SR_+(\theta + t/\sqrt{n}) \le N/2$. Under the true θ, $SR_+(\theta + t/\sqrt{n})$ has mean $N/2 - n^{3/2}t\int f^2 + o(n^{3/2})$ and variance $\sim n^3/12$, giving the limit $\Phi(t\sqrt{12}\int f^2)$. ■

F	ARE(sign, t)	ARE(SR, t)	ARE(sign, SR)
Uniform	$1/3$	1	$1/3$
Normal	$2/\pi \approx 0.64$	$3/\pi \approx 0.955$	$2/3$
Logistic	$\pi^2/12 \approx 0.82$	$\pi^2/9 \approx 1.10$	$3/4$
Laplace	2	1.5	$4/3$
Cauchy, slash	∞	∞	finite

(Check for $N(0,1)$: $f(0)^2 = \frac{1}{2\pi}$, $\int \phi^2 = \frac{1}{2\sqrt{\pi}}$. For Laplace $\frac{1}{2}e^{-|x|}$: $\sigma^2 = 2$, $f(0) = \frac{1}{2}$, $\int f^2 = \frac{1}{4}$.) Over all symmetric F, $ARE(SR, t) \ge 0.864$ (Hodges–Lehmann). So the signed-rank test costs little under normality and can gain a lot under heavy tails. The sign test wins only for very heavy tails.

2.4 Selected exercises

Motivation. The first exercise applies the sign-test machinery to a quantile other than the median; the second derives the tie-corrected moments of the signed-rank statistic that software uses for the normal approximation.

Exercise 2.4 (A sign test for the 10th percentile)

Ten foldable phones were folded until failure. The numbers of folds (in units of 10^4) were

$$7,\ 9,\ 12,\ 39,\ 28,\ 11,\ 22,\ 17,\ 14,\ 9.$$

Can one claim that at least 90% of such phones survive 10^5 folds? Use a sign-type test at level 0.1, and give a 90% confidence bound for the 10th percentile of the number of folds.

Solution.

Step 1. (Hypotheses.) Let the number of folds have a continuous distribution F and let $\xi = Q(0.1)$. "At least 90% survive 10 units" means $P(X < 10) \le 0.1$, i.e. $\xi \ge 10$. The claim must be the alternative: $H_0 : P(X < 10) = 0.1$ ($\xi = 10$) against $H_1 : P(X < 10) < 0.1$ ($\xi > 10$).

Step 2. (Test.) Let $K = \#\{i : X_i < 10\}$. The indicators $I(X_i < 10)$ are i.i.d. Bernoulli($P(X < 10)$), so under H_0, $K \sim B(10, 0.1)$, exactly as in Proposition 2.1 with $\frac{1}{2}$ replaced by 0.1. Small K supports H_1. The observed values below 10 are $7, 9, 9$, so $k = 3$ and

$$p = P(K \le 3) = 0.987.$$

H_0 is not rejected: three early failures out of ten are more than the one expected under H_0, so the data point the other way.

Step 3. (Confidence bound.) By the last paragraph of Section 2.1.1, $J = \#\{i : X_i \le \xi\} \sim B(10, 0.1)$. For continuous F there are a.s. no ties, and $\xi < X_{(u)}$ holds iff fewer than u observations are $\le \xi$, i.e. iff $J \le u - 1$. Since $P(J \le 1) = 0.736 < 0.9 \le 0.930 = P(J \le 2)$, the smallest valid choice is $u = 3$, and

$$P(\xi < X_{(3)}) = 0.930 \ge 0.9.$$

The ordered data are $7, 9, 9, 11, \ldots$, so $X_{(3)} = 9$. (The tie $9, 9$ comes from rounding and does not affect the order statistic used.) A two-sided interval $(X_{(l)}, X_{(u)})$ with $l \ge 1$ is impossible here: $P(\xi < X_{(1)}) = P(J = 0) = 0.9^{10} = 0.349$ already exceeds 0.05.

Conclusion: with confidence at least 90% the 10th percentile is below 9×10^4 folds, so the claim is not supported. ■

Exercise 2.8 (Signed-rank moments with zeros and ties)

To test $H_0 : \theta = \theta_0$, let $D_i = X_i - \theta_0$. Rank all n values $|D_i|$ (zeros included, so they occupy the lowest ranks), using mid-ranks for ties, and let SR_+ be the sum of the ranks of the positive D_i. Suppose t_0 of the D_i are zero and the nonzero $|D_i|$ form tie groups of sizes $t_1, \ldots, t_g$ (an untied value is a group of size 1). For example $D = 0, 0, 1, 1, 2, 3, 3, 3$ has $n = 8$, $t_0 = 2$ and nonzero groups of sizes $2, 1, 3$. Show that under H_0 (symmetry of D about 0), conditionally on $|D_1|, \ldots, |D_n|$,

$$E(SR_+) = \frac{n(n+1) - t_0(t_0+1)}{4},$$
$$\mathrm{Var}(SR_+) = \frac{n(n+1)(2n+1) - t_0(t_0+1)(2t_0+1)}{24} - \frac{\sum_l (t_l^3 - t_l)}{48}.$$

Solution.

Step 1. (Signs given magnitudes.) Symmetry gives $P(D \in A) = P(D \in -A)$ for Borel A. For $A \subset (0, \infty)$ this says $P(D > 0, |D| \in A) = P(D < 0, |D| \in A)$, and the two events partition $\{|D| \in A\}$. Hence $P(D > 0 \mid |D|) = \frac{1}{2}$ on $\{|D| > 0\}$. By independence across i, given $(|D_1|, \ldots, |D_n|)$ the indicators $V_i = I(D_i > 0)$ for $i \in \mathcal{N} = \{i : D_i \ne 0\}$ are i.i.d. Bernoulli($\frac{1}{2}$), while $V_i = 0$ for $i \notin \mathcal{N}$. The ranks R_i are functions of the $|D_j|$, hence fixed under this conditioning, and $SR_+ = \sum_{i \in \mathcal{N}} V_i R_i$.

Step 2. (Mean.) $E(SR_+ \mid |D|) = \frac{1}{2} \sum_{i \in \mathcal{N}} R_i$. The zeros take positions $1, \ldots, t_0$ and the nonzero values positions $t_0 + 1, \ldots, n$. Mid-ranks do not change the sum over a tie group, so

$$\sum_{i \in \mathcal{N}} R_i = \sum_{k=t_0+1}^{n} k = \frac{n(n+1) - t_0(t_0+1)}{2}.$$

Step 3. (A tie group.) If a group occupies positions $a+1, \ldots, a+t$, all its members get $\bar{r} = a + \frac{t+1}{2}$, and

$$\sum_{k=a+1}^{a+t} k^2 - t\bar{r}^2 = \sum_{k=a+1}^{a+t} (k - \bar{r})^2 = \sum_{j=1}^{t} \left(j - \frac{t+1}{2}\right)^2 = \frac{t(t^2-1)}{12}.$$

Step 4. (Variance.) $\mathrm{Var}(SR_+ \mid |D|) = \frac{1}{4} \sum_{i \in \mathcal{N}} R_i^2$. By Step 3 applied to each nonzero group,

$$\sum_{i \in \mathcal{N}} R_i^2 = \sum_{k=t_0+1}^{n} k^2 - \sum_l \frac{t_l^3 - t_l}{12} = \frac{n(n+1)(2n+1) - t_0(t_0+1)(2t_0+1)}{6} - \sum_l \frac{t_l^3 - t_l}{12}.$$

Dividing by 4 gives the formula. With $t_0 = 0$ and no ties it reduces to Proposition 2.5.

Check on the example: the nonzero mid-ranks are $3.5, 3.5, 5, 7, 7, 7$, with sum 33 and sum of squares 196.5. So $E = 16.5 = (72 - 6)/4$ and $\mathrm{Var} = 49.125 = (1224 - 30)/24 - 30/48$. ■